

Subject Code: 01CT0704

Subject Name: Management Information System

B. Tech. Year – IV (Semester VII)

Objective:

The main goals of an MIS are to help executives of an organization make decisions that advance the organization's strategy and to implement the organizational structure and dynamics of the enterprise for the purpose of managing the organization in a better way for a competitive advantage.

Credits Earned: 03 Credits

Course Outcomes: After completion of this course, student will be able to:

1. Understand the role of management information systems in achieving competitive business advantage through informed decision making.
2. Analyze how information technology affects a firm in terms of value creation and brings strategic benefits to a firm.
3. Interpret how to use information technology to solve business problems.
4. Develop meaningful decision-making capacity for the purpose of acquisition, development, deployment and management of information systems.

Pre-requisite of course: NA

Teaching and Examination Scheme:

Teaching Scheme (Hours)			Credits	Theory Marks			Tutorial / Practical Marks		Total Marks
				E	I		V	T	
Theory	Tutorial	Practical		ESE	IA	CSE	Viva	Term Work	
03	00	00	03	50	30	20	00	00	100

Contents:

Unit	Topics	Hours
1	Organisations and Computing Introduction, Modern Organization-IT enabled- Networked-Dispersed- Knowledge Organization, Information Systems in Organisations- what are information systems?,	

	Brief history of computing- ENIAC: Way to commercial computers- Advent of artificial intelligence- advent of personal computing-Free Software Movement- Advent of Internet, The role of internet- Internet and Web: they are different-the internet changes everything. Strategic business use of IS: Interdependence between organization and IS, IS strategies for competitive advantage using Porter's Five Forces Model and Value Chain Model.	06
2	Managing Information Systems in Organisations Introduction, Managing in the Internet Age, Managing Information Systems in Organizations - IT Interaction Model, Challenges for Managers - Information to Build Information? -To spend them on information systems? -What capabilities should be built with information systems? -How should services be centralized? - What level of safety is required? What is the technology road map for the organization?	05
3	Competing with IT Introduction, competitive environment of business - Partnership for mutual benefit - Bargaining power of suppliers - Bargaining power of buyers and options - Barriers to entry-risk of industry regulations, IT, using competition to compete at low cost.	05
4	Decision Support Systems Introduction, Understanding DSS- MIS and DSS-Decision making-types of decisions, Analytics and Business Intelligence- BI techniques	04
5	Managing Data Resources Introduction, The Need for Data Management- History of data use, Challenges of Data Management- data independence- reduced data redundancy- data consistency- data access- data administration- managing concurrency-managing security- recovery from crashes-application development, Database Concepts- fields, records and files- basic architecture, Data Warehouses- data mining uses	06
6	Managing IT Function Introduction, challenges of managing IT function - modern IT environment - centralization vs. decentralization - IT security - technology selection, vendor management - vendor selection - vendor contracts and service levels – relationship management - vendor retention or termination	04
7	Ethical issues Introduction, key issues - privacy - workplace monitoring - power over users, information security: First Line of defence - People / Staff, Second Line of defence - Technology, Prevention, Detection and Response to Authority Contemporary / emerging technologies: Cloud and Mobile Computing, E-commerce, m-commerce, Internet of things.	08
8	Practical Students should emulate an organization and its processes and develop a hypothetical information system. Students should study information systems adapted by various business institutions.	04
Total Hours		42

Suggested Text books / Reference books:

1. Essentials of Management Information Systems by Kenneth Laudon, Jane Laudon PHI
2. Management Information systems by W.S. Jawadekar TMH
3. Information Technology for Management: Transforming Organizations in Digital Economy by Efraim Turban, Dorothy Leidner, Ephraim McLean and James Wetherby, Wiley
4. Information Systems: Managing the Digital Firm Management by Kenneth Laudon, Jane Laudon, Pearson

Suggested Theory distribution:

The suggested theory distribution as per Bloom's taxonomy is as per follows. This distribution serves as guidelines for teachers and students to achieve effective teaching-learning process.

Distribution of Theory for course delivery and evaluation					
Remember	Understand	Apply	Analyze	Evaluate	Create
10%	25%	25%	25%	10%	5%

Supplementary Resources:

1. <https://www.inc.com/encyclopedia/management-information-systems-mis.html>
2. <https://www.smartsheet.com/management-information-systems>